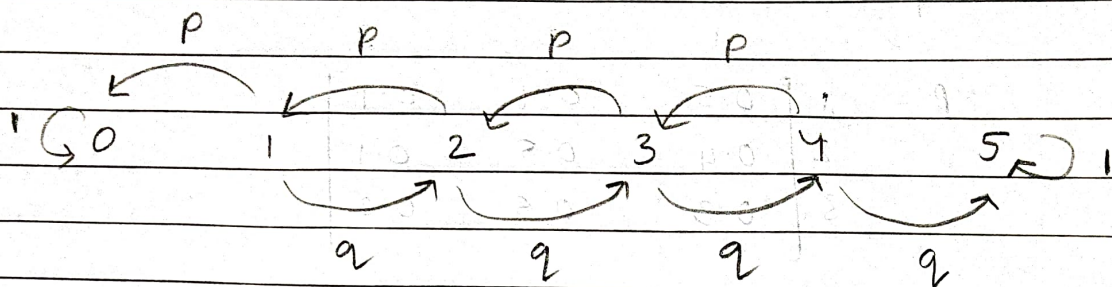


Assignment 9

1.



0 1 2 3 4 5

0	1	2	3	4	5
0	1	0	0	0	0
1	p	0	q	0	0
2	0	p	0	q	0
3	0	0	p	0	0
4	0	0	0	p	q
5	0	0	0	0	1

$$3. \quad P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix} \end{matrix}$$

$$(1) \quad P^{(0)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$$P^{(1)} = P^{(0)} \cdot P = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \cdot \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix} = \begin{bmatrix} 0.33 & 0.5 & 0.166 \end{bmatrix}$$

$$P^{(2)} = P^{(1)} \cdot P = \begin{bmatrix} 0.33 & 0.5 & 0.166 \end{bmatrix} \cdot P = \begin{bmatrix} 0.375 & 0.4582 & 0.166 \end{bmatrix}$$

$$P(X=2) = 0.166$$

2. 1 - pleasant

2 - bad

3 - fair

	1	2	3
1	0.5	0.4	0.1
2	0.4	0.5	0.1
3	0.3	0.5	0.2

 $X_0 = \text{Today}$ $X_1 = \text{Tmrw}$ $X_2 = \text{Day after tmrw}$

$$\rightarrow [X_2=1 | X_1=3] \cdot [X_1=3 | X_0=2]$$

$$(1) P(X_1=3, X_2=1 | X_0=2) = 0.1 \times 0.3 = \boxed{0.03}$$

$$(2) P(X_2=1 | X_0=2) \rightarrow [X_2=1, X_1=1 | X_0=2] + [X_2=1, X_1=2 | X_0=2] + [X_2=1, X_1=3 | X_0=2]$$

$$= (0.4 \times 0.5) + (0.5 \times 0.4) + (0.1 \times 0.3)$$

$$= \boxed{0.43}$$

$$3. (2) P(X_3=1, X_2=2, X_1=1, X_0=2) : (X_3=1 | X_2=2) \cdot (X_2=2 | X_1=1) \cdot (X_1=1 | X_0=2) \cdot P(X_0=2)$$

$$P^{(3)} = \cancel{P^{(2)}} \cdot p$$

$$= \frac{1}{3} \times \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4}$$

$$= \frac{3}{64} = \boxed{0.0468}$$

4.

0 = rains (rain)

1 = no rain

$$P = \begin{matrix} & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} a & 1-a \\ b & 1-b \end{bmatrix} \end{matrix}$$

where

 $a = 0.7$ $b = 0.4$

$$P = \begin{matrix} & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \end{matrix}$$

$$P(X_4 = 0, X_3 = 0, X_2 = 0, X_1 = 0 | X_0 = 0) :$$

$$P(X_0 = 0) \cdot P(X_1 = 0 | X_0 = 0) \cdot P(X_2 = 0 | X_1 = 0) \cdot P(X_3 = 0 | X_2 = 0) \cdot P(X_4 = 0 | X_3 = 0)$$

$$P^4 = \begin{matrix} & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{bmatrix} \end{matrix} \rightarrow 0.5749 = \boxed{57.49\%}$$

5.

R - Rain

N - No rain

$$\begin{aligned} 0 &\rightarrow (R, R) ; 2 \rightarrow (R, N) \\ 1 &\rightarrow (N, R) ; 3 \rightarrow (N, N) \end{aligned}$$

P =

$$P = \begin{matrix} & 0 & 1 & 2 & 3 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{bmatrix} \end{matrix}$$

 $X_0 = M$ $X_1 = T$ $X_2 = W$ $X_3 = Th$

$$P(X_3 = R | X_0 = R, X_1 = R) = ?$$

$$= P(X_2 = R | X_0 = R, X_1 = R) \cdot P(X_3 = R | X_1 = R, X_2 = R) + P(X_2 = N | X_0 = R, X_1 = R) \cdot P(X_3 = R | X_1 = R, X_2 = N)$$

$$= (0.7)(0.7) + (0.3)(0.4) = \boxed{0.61}$$

The model isn't a Markov Chain if the state is only 'today's weather' as it violates Markov's memoryless property.

In Markov Chain, the future must depend only on current state. Here, knowing it's raining today isn't enough; the probability of rain tomorrow changes depending on if it rained yesterday. So, we define a state as a two-day pair.